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AFRICAN 'SHADOW VOWELS': A DESCRIPTIVE SURVEY*

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From the time of the earliest descriptions of many East and Central African languages, one finds evidence for weakly articulated vowels referred to variously as 'semi-mute vowels', 'vowel-colored breaths', or 'shadow vowels'. The precise phonetic nature of these vowels -- which may be simply voiceless, or which may involve various phonation types -- remains unclear, and deserves further study. But apart from the intrinsic phonetic interest of these vowels, the available descriptions suggest phonological behavior that is also potentially of some importance. The Bongo-Bagirmi group of languages is particularly rich in phenomena involving shadow vowels, and these are described in some detail. Two patterns emerge: (i) in languages with both shadow vowels and normally articulated vowels on the surface (such as 'standard' Bagirmi), shadow vowels undergo processes (like deletion or assimilation) to which ordinary vowels are not susceptible; (ii) in languages with no surface shadow vowels (such as Ngambay, and some dialects of Bagirmi), certain vowels -- which, predictably, correspond with shadow vowels in related languages -- behave as if they were shadow vowels (i.e. as in (i) above). Finally, evidence of this second pattern (which raises interesting questions about the linguistic significance of such vowels in underlying representations) is presented from Alexandre's analysis of Bulu, in which apparently epenthetic 'buffer' vowels are in fact phonologically distinctive relics of the second vowel of original Bantu CVCV stems.

0. According to Tucker and Bryan (1966:60), 'An outstanding characteristic of the BONGO-BAGIRMI languages is that the final vowel of certain words is semi-mute, i.e. pronounced so softly as to be hardly audible, and readily elided before following sounds'. This is a phenomenon which has been documented since the earliest work on Bongo-Bagirmi, as can be seen in such descriptions as the following:

(1a) Delafosse 1897:19 (for Sara):

> Souvent en sara les voyelles finales sont excessivement brèves; on pourrait même ne pas les prononcer, et elles disparaissent parfois, dans les mots composés; comme ce n'est pas constant, et que beaucoup de voyelles finales conservent au contraire toute leur valeur, j'ai distingué les voyelles semi-muettes par le signe ◡ qui marque la brièveté. Ainsi les mots debĕ homme, monŏ enfant, devront se prononcer presque deb, mon, en faisant à peine sentir la voyelle finale. Au pluriel on pourrait dire et écrire: deb ge, mon ge.

Compare Gaden 1909:8 (for Bagirmi):

> (1b) Le signe / placé en avant d'une voyelle, indique que cette voyelle disparaît dès que le mot n'est pas isolé ou en finale. Exemple:
> kag/a, arbre; kag muta, trois arbres.
> Quand ce signe est absent, le mot est invariable:
> kaga, panthère; kaga muta, trois panthères.

More recently, cf. Caprile 1968:9 (for Mbay):

> (1c) En position faible ... la réduction des oppositions vocaliques est très importante. Le relâchement, la centralisation, la brièveté et jusqu'à un dévoisement partiel caractérisent les réalisations vocaliques dans cette position et rendent leur identification assez difficile ...

But this phenomenon is not peculiar to the Bongo-Bagirmi languages: it has been observed in a variety of East and Central African languages, such as Teso, Burunge, and Galla, with documentation, again, from some of the earliest descriptions. For Galla, witness the following statement by Praetorius (1893:4):

> (2) Alle auslautenden unbetonten Vokale neigen zum Abfall. Sie können entweder 'ganz stummen sein, oder doch so kurz gesprochen werden, dass sie nur noch ein Hauch zu sein scheinen, immerhin aber so, dass der Vokal noch kennbar ist'.

The internal citation ('ganz stummen sein ... noch kennbar ist') indicates, in fact, Praetorius' debt to still earlier sources (in this case a previous description by Tutschek.)

The cumulative impression left by informal descriptions such as these, as well as Tucker's more recent statement that the semi-mute vowels of Galla are 'voiceless and barely articulated' (1967:661), is that we have to do with phonetically voiceless vowels; compare also the traditional designations 'semi-mute vowel' and 'vowel-colored breath'. But precise phonetic data on such Vs (including instrumental analyses) are so far lacking, and it seems preferable to adopt, at least provisionally, the phonetically neutral term 'shadow vowel' (due to Hilders and Lawrance 1956). Despite the lack of such phonetic data, however, I would like to suggest that the phonological behavior of these Vs, as seen in the available published descriptions of the languages in question, is potentially of great interest[1].

The paper examines, then, the following areas: first, the behavior of shadow vowels (henceforth V̆s) in synchronic phonological systems, particularly in Bagirmi; second, the behavior of Bongo-Bagirmi V̆s from a historical/comparative point of view; and third, a brief comparison of Bongo-Bagirmi V̆s with similar data elsewhere in Africa. The aim of the paper is not so much to propose and defend 'unique' or 'correct' analyses of the data, but

rather to characterize the potential interest and importance of these neglected phenomena, if only as a stimulus for future work, both descriptive and theoretical.

1. In Bagirmi, Vs are susceptible to certain phonological processes which do not apply to normally articulated Vs. First, Vs (the best examples involve /a/) assimilate to bilabial or palatal nasal consonants of following morphemes, as can be seen in the following alternations:

(3) /aka/ 'see': m-aka 'I see, saw' j-akʊ-ma 'they saw me'
j-aki-ña 'they saw him'

/ada/ 'give': m-ada 'I give, gave' n-ádù-m gà 'he gave me'
m-ádí-ñ gà 'I gave him'

/taʃa/ 'work, do': ma m-taʃa 'I do, did' tádí-ñ 'do it!'

Compare /aca/ 'cut', which does not have a V:

(4) /aca/ 'cut': m-aca 'I cut' m-ácá-ña 'I cut it' (etc.)

Before non-nasal consonants of following morphemes, and before the dental nasal [n], Vs are deleted; compare the following forms from the paradigm of /ada/ 'give' with the assimilated forms n-ádù-m gà 'he gave me' and m-ádí-ñ gà 'I gave him' in (3) above:

(5) /ada/ 'give': n-áj-jè 'he gave us'
n-át-sè 'he gave you (pl.)'[2]

Some further examples are provided in (6) (note the morpheme-initial [n] in the first item; morpheme-initial [ŋ] does not occur):

(6) manɔ 'head of cattle' → maŋ-ne 'cow'
bɪsi 'dog' bɪs-ge (bɪz-ge) 'dogs'
kágá 'tree' kag-ge 'trees'
táʃá 'work, do' táʃ-kí 'work! (pl.)'
ŋáŋá 'tooth' ŋáŋ-la 'chew'
abe 'go, leave' ab-se 'carry off'

V-deletion is in fact more general, and applies across word-boundary (including compound-boundary), even if the following word begins with a bilabial nasal:

(7) kamo̰ 'eye' → kám kàgá̰ 'leaf' (lit. 'eye tree')
tòtò̰ 'hill' tòt mbásá 'stone' (lit. 'hill small')
ŋabà̰ 'man' ŋab ga 'man who'
tádá̰ 'work, do' m-tád̰ gà 'I have done'
bogo̰ 'steal' mala bog ja 'the master (who) stole the meat'
kágá̰ kág mùtá '3 trees'

(There are no examples before the palatal nasal [ñ], which does not occur word-initially.)

In contrast, ordinary Vs do not delete before following morpheme- or word-initial consonants (cf. also (4) above):

(8) kàgà 'leopard' → kaga-ge 'leopards'
ŋágla 'chewed' ŋágla-ki 'chewed (pl.)'
ali (n. prop.) ali-ge 'people named Ali'
màrà 'crocodile' màrà ngólò 'big crocodile'
gwoto 'be absent' ne gwoto lolo̰ 'he is not here'

In Sara and Kenga, preconsonantal V̰s behave somewhat differently: according to Tucker and Bryan (1966:63), the preconsonantal V̰ deletes, as in Bagirmi, but this leads to a further development, namely the insertion of a syllabic nasal consonant, homorganic to the preceding consonant:

(9) Sara: bɔ́bà̰ 'male' → bɔ́bm̩-maŋ 'bull' ('male-cow')
Kenga: màkà̰ 'belly' da-màkŋ̩-kabà̰ 'carnivore' ('meat-belly-wild animal')
kɔsɔ̰ 'food, eat' kɔsn̩ kɔ̀s 'to eat food'

(Both Vs and V̰s generally elide before following Vs.)

A somewhat different situation is found in other dialects of Bagirmi, as can be seen from field work of J. Lukas, as reported in Tucker and Bryan. In the material designated 'Lukas 1' in Tucker and Bryan's corpus, from a single village, no V̰s are recorded for many words which consistently have a V̰ elsewhere in Bagirmi; these forms exhibit, instead, normally articulated Vs, as in the forms in (10):

(10)	'standard' Bagirmi	Lukas 1
	kágá̰ 'tree'	kágá
	ʔaŋa̰ 'sheep'	àŋá
	ímḭ́ 'locust'	ìmí
	ɪjɪ̰ 'urine'	ɪ́jí
	mùjṵ́	mùjú

ʔʌmu$_{\chi}$ 'nose'	àmú
ulu$_{\chi}$ 'swallow'	ùlú
élẹ$_{\chi}$ 'bird'	èlé
dɛbẹ$_{\chi}$ 'person'	dèɓé
k-abẹ$_{\chi}$ 'journey'	k-àɓe
manɲo$_{\chi}$ 'head of cattle'	màɲó
oso$_{\chi}$ 'bite'	ósó
ʔabo$_{\chi}$ 'hippo'	àfó
ɲono$_{\chi}$ 'child'	ɲónó

Nevertheless, such 'Lukas 1' forms undergo the same processes described for Vs above, while 'Lukas 1' forms which correspond to Vs elsewhere do not. For assimilation, consider the form 'bite' illustrated in (11):

(11) Lukas 1

ìñí 'snake' + oso 'bite' : ìñ ósú-má 'the snake bit me'
ìñ ósí-ñà 'the snake bit him'

For deletion, consider the forms in (12):

(12) Lukas 1

/ìñí#òsó+jè/	→	ìñ òs-jè 'the snake bit us'
màɲó 'head of cattle'		màɲ-né 'cow'
dèɓé 'person'		dèɓ-gé 'people'
ùlú 'swallow'		m-úl gà 'I swallowed'

In the dialect of still another village, designated 'Lukas 2', final Vs of any kind are altogether missing, as can be seen from forms like those in (13) (the corresponding forms from (10) above are repeated for convenience):

(13) 'standard' Bagirmi	Lukas 1	Lukas 2
kágá$_{\chi}$ 'tree'	kágá	kák
ʔaɲa$_{\chi}$ 'sheep'	àɲá	aŋ
ími$_{\chi}$ 'locust'	ìmí	ìm
ʔʌmu$_{\chi}$ 'nose'	àmú	ʌm
élẹ$_{\chi}$ 'bird'	èlé	èl
dɛbẹ$_{\chi}$ 'person'	dèɓé	dèɓ
manɲo$_{\chi}$ 'head of cattle'	màɲó	màɲ
ʔabo$_{\chi}$ 'hippo'	àfó	àp
ɲono$_{\chi}$ 'child'	ɲónó	ɲón

Descriptively, then, we find in Bagirmi the following three synchronic situations:

(i) In the dialect of what may be called 'standard' Bagirmi, $\underset{x}{V}$s are phonetically distinct, on the surface, from other Vs.[3] The fact that $\underset{x}{V}$s are distributionally restricted, however, could be taken as an indication that such Vs are underlyingly not distinct from other Vs, and result from some low-level process (whether optional, obligatory, or 'variable') which applies to Vs in word-final position. One might claim, for example, that phonetic $\underset{x}{V}$s arise automatically in unstressed word-final position, while final stressed Vs remain unaffected: thus Bag. ['kágá] 'tree' versus [kà'gà] 'leopard'. It appears, however, that in the Bongo-Bagirmi languages generally, 'stress is normally on the first syllable of disyllabic words, and invariably so when the second syllable contains a semi-mute vowel' (Tucker and Bryan 1966:64). Thus, while it is probably true that $\underset{x}{V}$s are never stressed, most other final Vs are likewise generally unstressed, resulting in actual surface contrasts like Bag. ['kágá] 'tree' versus ['kàgà] 'leopard'. Apart from stress (which seems, moreover, to have no linguistic significance whatsoever in Bongo-Bagirmi), there are no other known factors which might condition the realization of Vs as $\underset{x}{V}$s.[4] In similar fashion, one could claim that $\underset{x}{V}$s are automatically inserted, under certain conditions. There are two immediate problems for such an analysis: first, while the vowel quality of $\underset{x}{V}$s is identical to that of the V of the preceding syllable in a large number of cases, such that one might wish to envisage (e.g.) a copying relationship, this is by no means a general fact; second, the tones appearing on $\underset{x}{V}$s do not appear to be predictably related to any word-level features, including the nature of the preceding consonant. But the major obstacle to this and similar analyses is the fact that $\underset{x}{V}$s regularly undergo certain phonological processes which normal Vs do not. As an alternative, it may be necessary to consider $\underset{x}{V}$s as underlyingly distinct from other Vs, i.e. specified by some (perhaps laryngeal) feature or features, depending on their precise phonetic realization. This sort of situation, while typologically somewhat rare, is not excluded (given, e.g., the existence of systems with distinctive laryngealized or murmured Vs); one may further compare, in fact, the nasalized vowels of the Sara languages (a subgroup of the Bongo-Bagirmi languages), which must be taken to be underlying (i.e. not derived from sequences of oral vowel plus nasal consonant), a situation which I have discussed elsewhere (1978).

(ii) In contrast with the 'standard' Bagirmi situation outlined in (i), the dialect referred to as 'Lukas 1' shows a more complex situation. Both $\underset{x}{V}$s and certain normal Vs (the latter corresponding to $\underset{x}{V}$s in 'standard' Bagirmi) undergo those processes restricted to $\underset{x}{V}$s in 'standard' Bagirmi. It is difficult to know how to account for such a situation. On the face of it, one is tempted to claim that the state of affairs in 'Lukas 1' is an artifact of faulty description: perhaps the forms with apparent V for $\underset{x}{V}$ (like those in (10) above) really have phonetic $\underset{x}{V}$s, in which case 'Lukas 1' is identical with 'standard' Bagirmi. But there is evidence, as we shall see, in favor of taking the description of 'Lukas 1' at face value -- i.e., evidence that a functional equivalence, as it were, of Vs and $\underset{x}{V}$s is possible, and linguistically significant.

(iii) Finally, $\underset{x}{V}$s are altogether absent in the dialect referred to as 'Lukas 2', posing no particular problems for the synchronic analysis, but indicating certain interesting consequences for the comparative and diachronic analysis of $\underset{x}{V}$s in the Bongo-Bagirmi languages, to which we now turn.

2. From a comparative perspective, the Bongo-Bagirmi languages as a whole pattern, with respect to $\underset{x}{V}$s, much like the dialect situation internal to Bagirmi, as described above. Thus, some languages contain $\underset{x}{V}$s which behave much like those in 'standard' Bagirmi (e.g. Bongo, Baka, Yulu, some dialects of Sara, etc.; cf. n. 3), while others appear to contain no reflexes of $\underset{x}{V}$s at all, as in 'Lukas 2' (e.g. most of the dialects of 'Sara proper', i.e. Tucker and Bryan's 'Sara Mbai'). But most interesting are those languages comparable to 'Lukas 1', in which $\underset{x}{V}$s are phonetically nonexistent, but in which certain phonological effects must be interpreted as traces of $\underset{x}{V}$s.

In Ngambay (cf. Vandame 1963, Thayer and Thayer 1971), no $\underset{x}{V}$s are recorded. Nonetheless, certain final Vs behave like $\underset{x}{V}$s, from the standpoint of e.g. Bagirmi. Consider first certain facts involving word-final /-ə/ in Ngambay.

According to Vandame (1963:13), it is sometimes difficult to distinguish a real word-final phonemic /-ə/ from an automatic stop-release, particularly when the [-ə] bears the same tone as the V of the preceding syllable. He proposes to make the distinction on the basis of the behavior of [-ə] before following Vs, as follows: prevocalic V-elision is not obligatory, especially in deliberate speech, and yet some [-ə]s consistently elide before Vs, while others consistently do not. Those words in which [-ə] always remains on the surface are taken to contain /-ə/, while the [-ə] which consistently elides in other words is taken to represent a stop-release. Thus, the words for 'grain' and 'tree' show identical preconsonantal realizations, with respect to word-final [-ə]:

(14) [kándə̀ tò ɗá] 'The grain is where?'
[kāgə̀ tò ɗá] 'The tree is where?'

But their prevocalic realizations differ, as follows:

(15) [kándə́ ɔ̀sō nàŋ] 'The grain fell to the ground.'
[kāg ɔ̀sō nàŋ] 'The tree fell to the ground.'

Thus 'grain' is analyzed as /kándɔ́/, while 'tree' is analyzed as /kāg/, with a stop-release appearing in preconsonantal position, as in (14).

But this analysis is somewhat misleading, once comparative evidence is considered. Vandame does not seem to have noticed that the word for 'tree', for example, corresponds to forms with $\underset{x}{V}$ elsewhere in Bongo-Bagirmi, thus Bagirmi kágá, Bongo kaga, Yulu kage.[5] We are dealing, then, in all likelihood, not with a simple stop-release, but rather with a phonological property of certain historical $\underset{x}{V}$s, which surfaces in some environments. Vandame's analysis, if synchronically correct, would thus be the result of a restructuring, whereby forms like *kāgə, with -$\underset{x}{V}$, together with a prevocalic $\underset{x}{V}$-dele-

tion rule, were reanalyzed as /kāg/, with a preconsonantal ə-insertion rule. But it is at least worth considering the more abstract analysis, whereby surface [-ə#] < *-ə and surface [-ə#] < *-$\overset{x}{ə}$ are still underlyingly distinct, comparable to the situation in the 'Lukas I' dialect of Bagirmi, as outlined above.

A second indication of the linguistic significance of $\overset{x}{V}$s in languages which do not distinguish them phonetically concerns certain word-final tone-bearing sonorants in Ngambay. According to Vandame (1963:37-8), liquids and nasals in initial and intervocalic position automatically bear the tone of the following V. In word-final position, however, certain liquids and nasals bear unexpected tones. Vandame attempts to account for these by invoking sentence-sandhi (i.e. influence, in this case, from following V-initial words) or some sort of conditioned influence from preceding Vs. Unfortunately, Vandame does not provide sufficient data to justify a detailed analysis at this time. There is reason to believe, however, that the explanation for such 'unexpected' tones -- once again, from a historical point of view -- is that they reflect traces of $\overset{x}{V}$s. Thus Ngambay /màn/ 'water', realized either as [màǹ] or [màn̄], is not to be separated from Bagirmi mane, Bongo mini, Yulu mimi, etc. Essentially the same process is quite transparent in the Ngambay first person suffixed pronoun, the only case, significantly, in which a tone-bearing sonorant is distinctive: thus Ngambay -$\overset{x}{m}$ 'me' corresponds to Bagirmi -$\overset{x}{m}$á. (Note also that in this way, it is possible to make some sense of Tucker and Bryan's remark (1966:64) that in those dialects of Sara with phonetic $\overset{x}{V}$s, no $\overset{x}{V}$s are found after liquids and nasals.)

3. Turning now to languages outside of the Bongo-Bagirmi group: there are indications that $\overset{x}{V}$s represent an interesting 'areal' feature of sorts, which appears in several diverse forms.

Very little material is available concerning Vs described as $\overset{x}{V}$s throughout the rest of the linguistic area covered by Tucker and Bryan, namely the $\overset{x}{V}$s asserted to exist in Burunge, Geleba, Krongo, the Teda-Tubu dialects, Shilluk, Teso and Galla. The available evidence, however, indicates that $\overset{x}{V}$s in these languages have essentially the same properties as $\overset{x}{V}$s in the Bongo-Bagirmi languages, namely (i) they are restricted to word-final position; (ii) there are no restrictions as to which consonants may precede them; (iii) they may be tonally distinct from preceding Vs in the word, although they are often identical in quality to the V of the preceding syllable; (iv) they tend to elide more readily than normal Vs; (v) they appear, in related languages or dialects, either as zero or as normal Vs.[6] Although $\overset{x}{V}$s are evidently not phonologically distinct in certain of these languages, minimal contrasts like those in (16) indicate that they may somehow be distinct at least in Galla:

(16)	'person'	acc. nāmà$\overset{x}{}$	dat. nāmà (m.)
	'girl'	acc. íntālà$\overset{x}{}$	dat. íntālà (f.)
	'this/these'	acc. káná$\overset{x}{}$	gen. káná (m.)

But more interesting, from a broader, typological perspective, are languages with no trace of phonetic Vs, but which present phonological behavior reminiscent of Vs, and which should perhaps be further investigated in this light. I would like to suggest that such a case is represented by Bulu, a Bantu language spoken in southern Cameroun, and thus roughly adjacent to the southernmost extensions of Bongo-Bagirmi territory. The facts in question have been discussed by Alexandre (1962), on which the following summary presentation is based.

Several groups of Bantu languages of southern Cameroun are characterized by a relatively high frequency of monosyllabic stems ending in a closed syllable, corresponding to CVCV stems elsewhere. Typical languages of this sort are Mbɛnɛ, Makaa, and Bulu. But, while e.g. Mbɛnɛ and Makaa present, as a result, a high frequency of consonant clusters in actual speech, this is not the case in Bulu; consider the following descriptions (references in Alexandre 1962):

(17) 'A word ending in a consonant, when there follows another word beginning with a consonant, requires a slight vowel sound after it, making a little buffer syllable between the two words.' (Bates)

'When a word ends in a consonant, closely followed by another word beginning with a consonant, Bulu speech very commonly inserts a slight vowel sound between.' (Good)

Alexandre, however, has shown that this apparently anaptyctic 'buffer' V is not the result of a low-level epenthesis, but a reflection of a segment which is phonologically distinctive. While the quality of these 'buffer' Vs in Bulu is nearly always identical to, or systematically related to, the quality of the preceding V, their tones bear no such relationship. Moreover, the tonal patterns of these Vs are functionally identical to those of the second V in CV_1CV_2 stems. Finally, Alexandre suggests that the Bulu situation may represent an intermediate stage in a gradual attrition process affecting final Vs, with the end stage, namely zero, represented in languages like Mbɛnɛ or Makaa.

This sort of situation is strongly reminiscent of the patterns outlined for Vs above, both with respect to their surface properties and distribution, and with respect to their phonological behavior, as exemplified in particular by the analysis of the so-called 'epenthetic' [-ə#] of Ngambay, involving three-way contrasts like Bagirmi kágá, Ngambay (preconsonantal) [kāgə], Sara kág 'tree'.

NOTES

*This is a slightly revised version of a paper presented at the 10th Annual Conference on African Linguistics, held at the University of Illinois at Urbana-Champaign in April of 1979. An expanded version appeared in 1981 (see the REFERENCES), to which the reader is referred for extensive discussion of some of the theoretical issues raised by the data.

[1]In addition to the data in Tucker and Bryan 1966:59ff., material from the Bongo-Bagirmi languages has been drawn from the following sources: Gaden 1909 (Bagirmi); Hallaire and Robinne 1959 (Sara); Caprile 1968 (Mbay); Vandame 1963, Thayer and Thayer 1971 (Ngambay); Santandrea 1963 (Bongo, Baka, Yulu, Kara). Tone is marked where known. For a more complete description of these and other relevant sources, see Vine 1978:119-20.

[2]The regressive assimilations exhibited by such forms are optional, or rather 'variable', and depend on such factors as speech style, speed of utterance, etc. For a similar situation, see Bolozky 1980:797 with respect to regressive voicing assimilation in Hebrew.

[3]The same situation is found in several other Bongo-Bagirmi languages, including Bongo, Baka, Yulu, Kenga.

[4]Tucker and Bryan (loc. cit.) note that 'occasionally stress is heard on the second syllable, when the first one contains a central vowel', while citing as an example Stevenson's description of Bag. kə̀lá 'snake' as phonetically [kə̀'lá]. Such forms would constitute, then, the only class of exceptions to the claim that stress in disyllabic words is initial. But one can also question the status of this class as 'exceptional', since the overwhelming majority of such CəCV forms (extremely frequent, for example, in Ngambay) have high tone on the final V. One wonders, then, whether the schwa of the first syllable (which characteristically has non-high pitch) and the concomitant high tone of the second syllable combine to give an impression of stress, which need not actually be present. It should be added that the distribution of /ə/ (as seen particularly in Ngambay) is by no means restricted to this position (Cə̀CV́); therefore, it would be problematic to claim, without further justification, that this /ə/ actually constitutes a 'reduction' of some full V, conditioned, in fact, by word-final stress. The problems associated with the status of /ə/ in these languages, however, are complex, and cannot be considered further here.

[5]It is along these lines that certain Ngambay forms in Thayer and Thayer's material, with tone marked on word-final obstruents, are probably to be understood, e.g. kã̄ǵ 'tree'. A number of such forms are consistently marked in this way by Thayer and Thayer, despite their claim that only Vs and nasals bear tone (1971:3.24). Thus Bag. mosu, Sara (dial.) masę, but Thayer and Thayer mɛ́s̄ 'blood' for Ngambay; Bag. kese, but Thayer and Thayer kɛ́s̄ 'cough', etc.

[6]With respect to the last feature: an interesting situation worthy of further investigation concerns the Vs in certain Nilotic languages. In Shilluk, most verbs have the shape CVCV̂, whereas in (e.g.) Acoli, the corresponding verbs are of the shape CVCV, with a normally articulated final V. Nevertheless, this final V, which has no lexical value, seems also to be phonologically 'irrelevant': it never appears on the surface if a form undergoes any phonological process whatsoever, and may not belong in underlying representations at all. For details, see Crazzolara 1955.

As a postscript, let me report the existence of independent work on issues very closely related to those treated above. In a letter dated July 14, 1980, Gerrit Dimmendaal informs me that his paper 'Non-voiced vowels in Turkana: a Nilo-Saharan feature?' (in Collected Seminar Papers 1979/80, ed. Franz Rottland: Dept. of Linguistics and African Languages, Univ. of Nairobi) argues that voiceless Vs in Turkana have phonemic status, on the basis of data concerning such factors as stress and tonal patterning. He notes, in addition, that his previous work on certain Nilotic languages (among them Teso and Shilluk, cf. sections 0. and 3. above, with n. 6) has led to similar conclusions. On the question of the phonemic status of voiceless Vs in a number of languages, see also my earlier discussion (1981:401-8).

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